# Lab 3: GEDA + Analytic Questions

Visualization Methods in Data Analysis by R

Group 4, Variant 5

2026-03-09

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### Dataset: Palmer Penguins

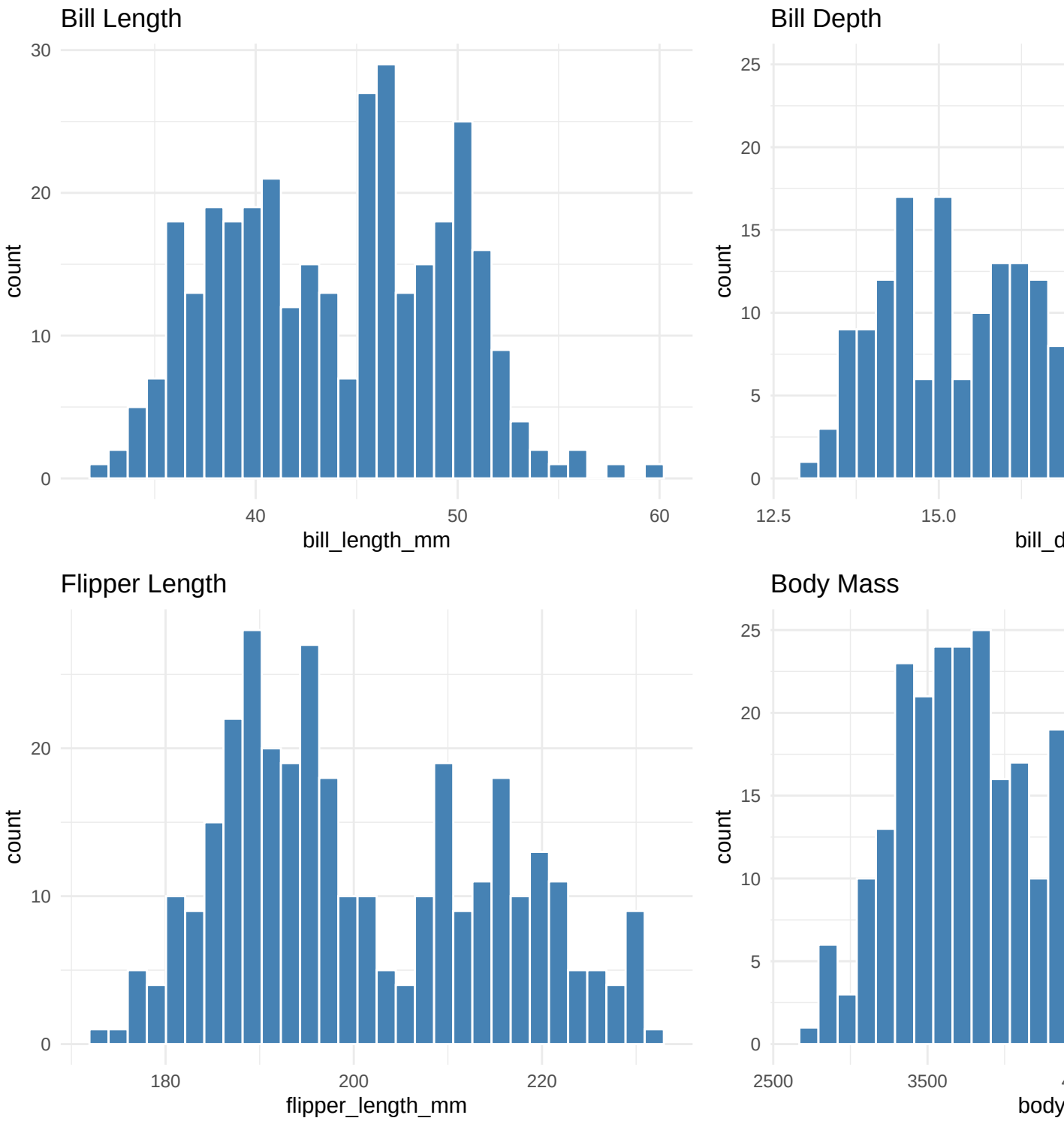
```r
glimpse(penguins_clean)
```

```
## Rows: 333
## Columns: 8
## $ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
## $ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
## $ bill_length_mm    <dbl> 39.1, 39.5, 40.3, 36.7, 39.3, 38.9, 39.2, 41.1, 38.6~
## $ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, 19.3, 20.6, 17.8, 19.6, 17.6, 21.2~
## $ flipper_length_mm <int> 181, 186, 195, 193, 190, 181, 195, 182, 191, 198, 18~
## $ body_mass_g       <int> 3750, 3800, 3250, 3450, 3650, 3625, 4675, 3200, 3800~
## $ sex               <fct> male, female, female, female, male, female, male, fe~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

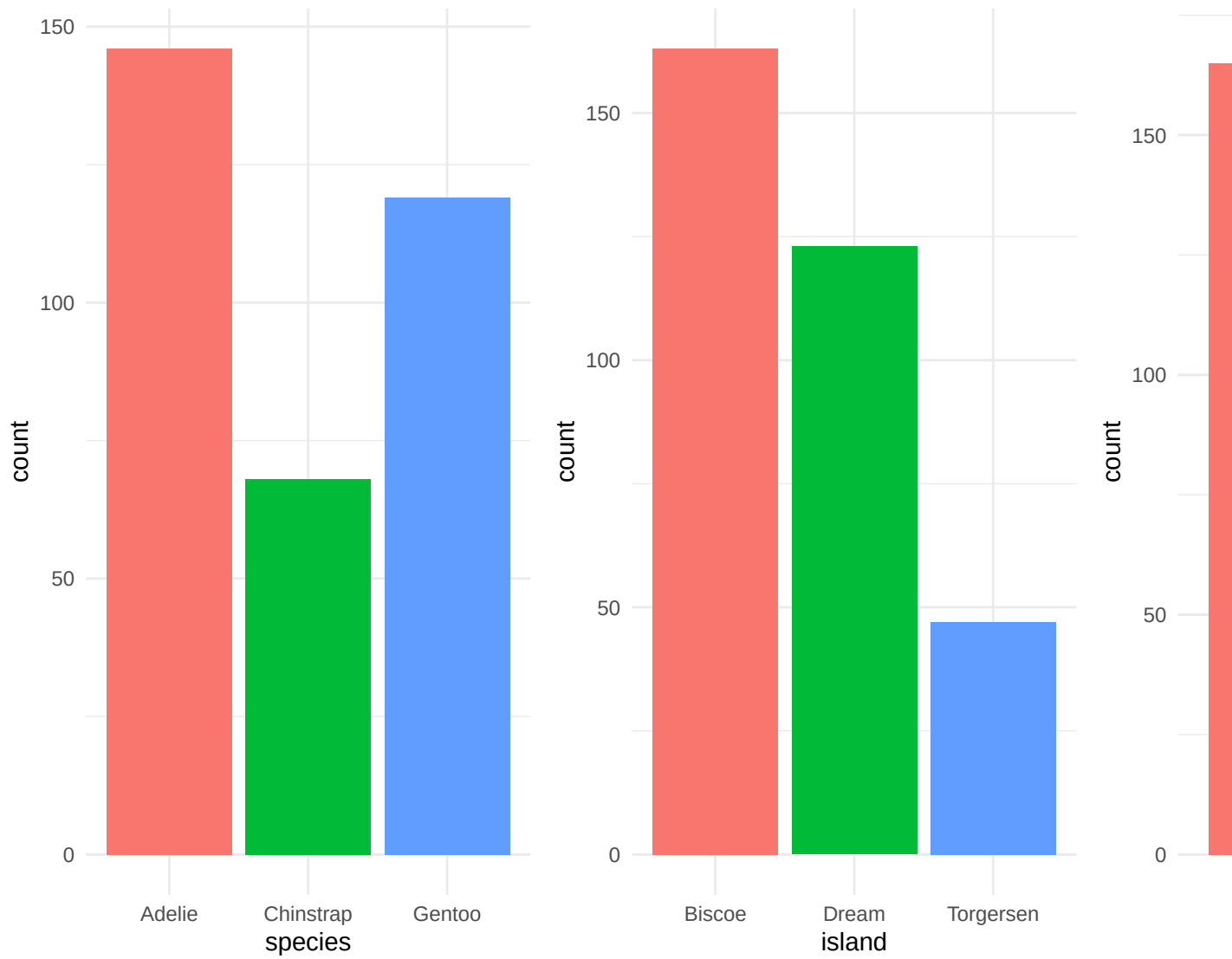
### GEDA: Distribution Overview

```r
p_bill_len <- ggplot(penguins_clean, aes(bill_length_mm)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 30) +
  theme_minimal() + labs(title = "Bill Length")

p_bill_dep <- ggplot(penguins_clean, aes(bill_depth_mm)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 30) +
  theme_minimal() + labs(title = "Bill Depth")

p_flip <- ggplot(penguins_clean, aes(flipper_length_mm)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 30) +
  theme_minimal() + labs(title = "Flipper Length")

p_mass <- ggplot(penguins_clean, aes(body_mass_g)) +
  geom_histogram(fill = "steelblue", color = "white", bins = 30) +
  theme_minimal() + labs(title = "Body Mass")
```

```r
(p_bill_len + p_bill_dep) / (p_flip + p_mass)
```



```r
p_species <- ggplot(penguins_clean, aes(species, fill = species)) +
  geom_bar(show.legend = FALSE) + theme_minimal()
p_island <- ggplot(penguins_clean, aes(island, fill = island)) +
  geom_bar(show.legend = FALSE) + theme_minimal()
p_sex <- ggplot(penguins_clean, aes(sex, fill = sex)) +
  geom_bar(show.legend = FALSE) + theme_minimal()

p_species + p_island + p_sex
```

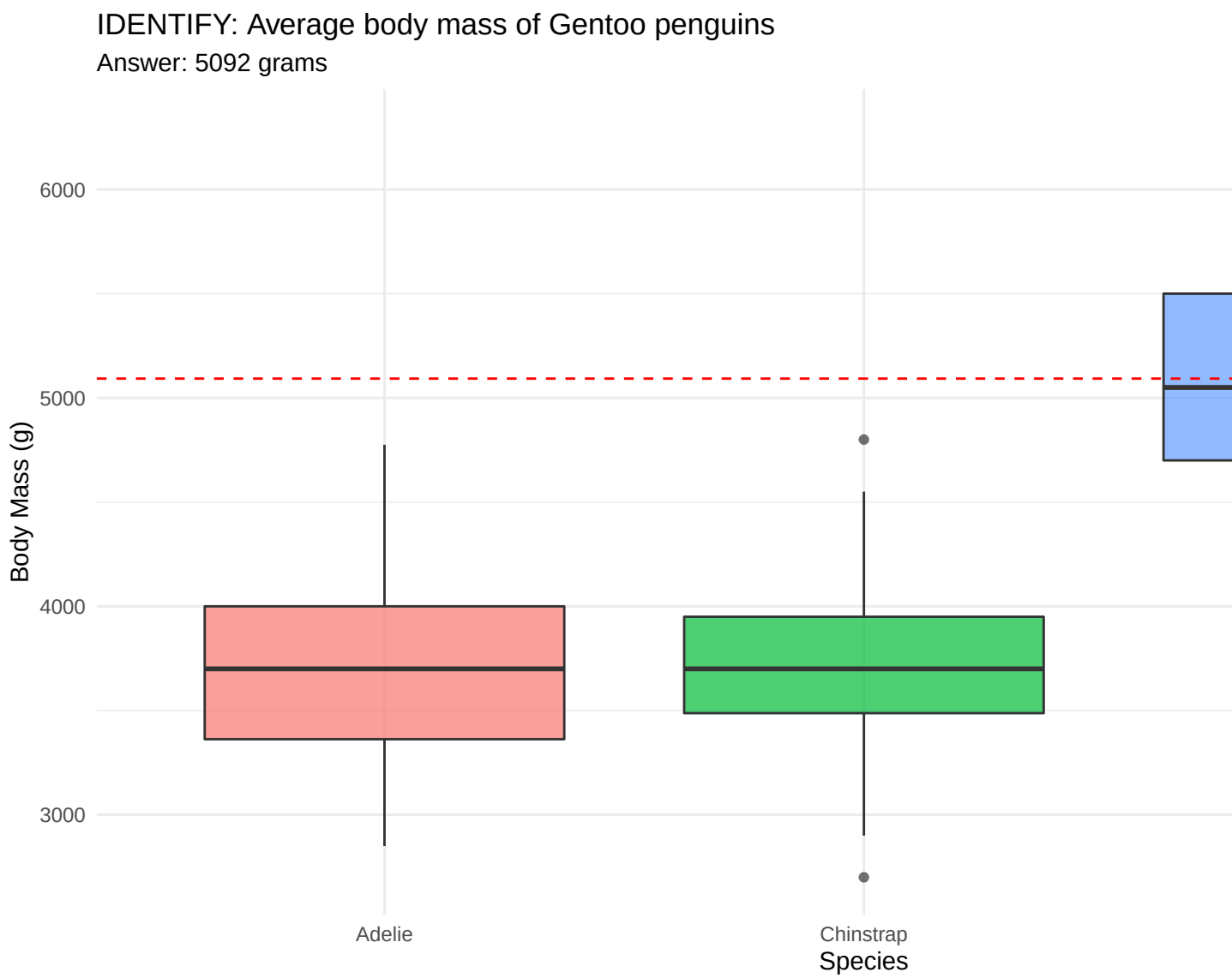



## Elementary Questions

### 1. IDENTIFY: Average body mass of Gentoo penguins?

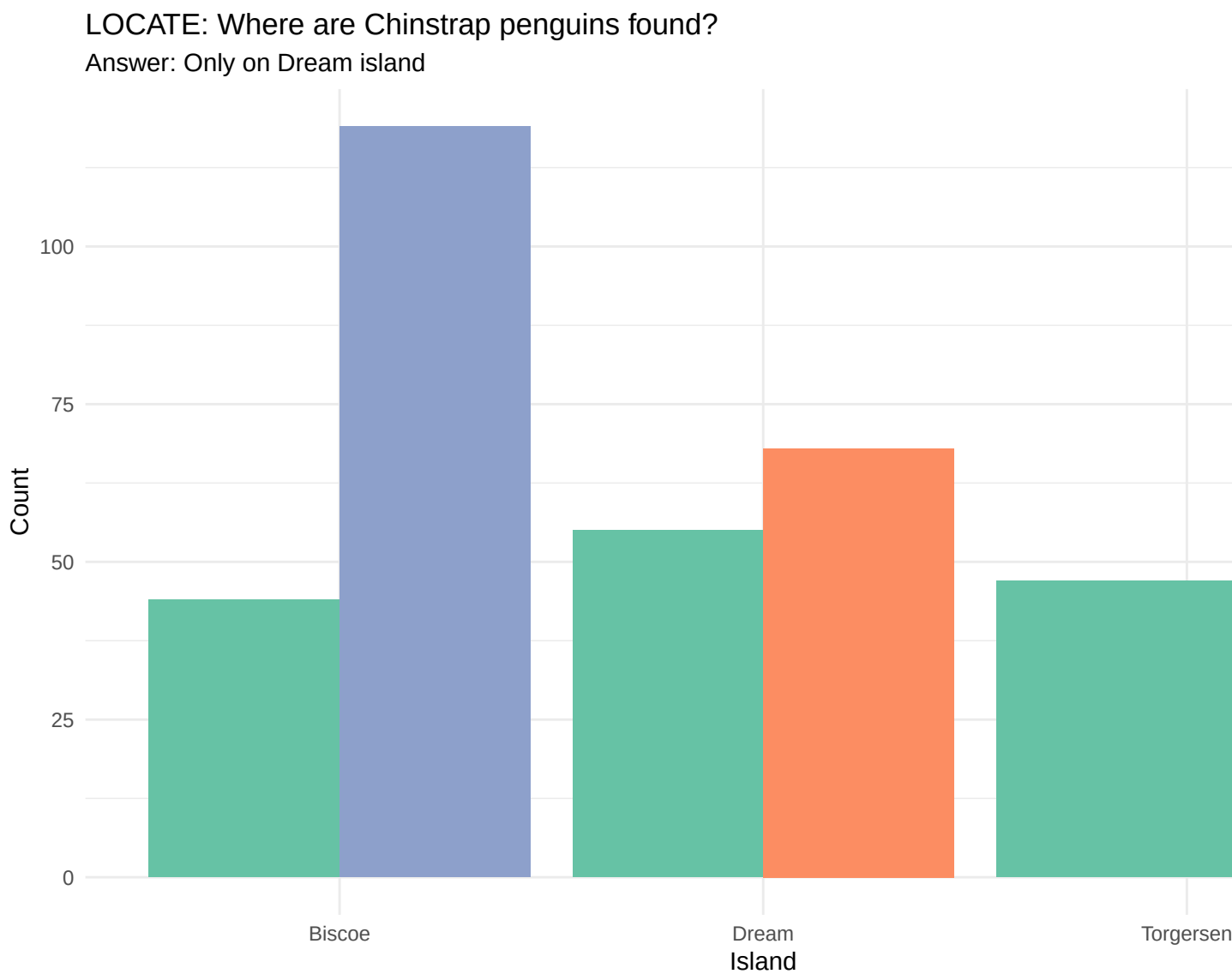
```r
gentoo_mass <- penguins_clean |>
  filter(species == "Gentoo") |>
```

```r
  summarise(mean_mass = mean(body_mass_g))

ggplot(penguins_clean, aes(x = species, y = body_mass_g, fill = species)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.7) +
  geom_hline(yintercept = gentoo_mass$mean_mass, linetype = "dashed", color = "red") +
  annotate("text", x = 3.3, y = gentoo_mass$mean_mass + 100,
           label = paste0("Gentoo mean: ", round(gentoo_mass$mean_mass), "g"),
           color = "red", size = 3.5) +
  labs(title = "IDENTIFY: Average body mass of Gentoo penguins",
       subtitle = paste0("Answer: ", round(gentoo_mass$mean_mass), " grams"),
       x = "Species", y = "Body Mass (g)") +
  theme_minimal()
```



**2. LOCATE: Where are Chinstrap penguins found?**

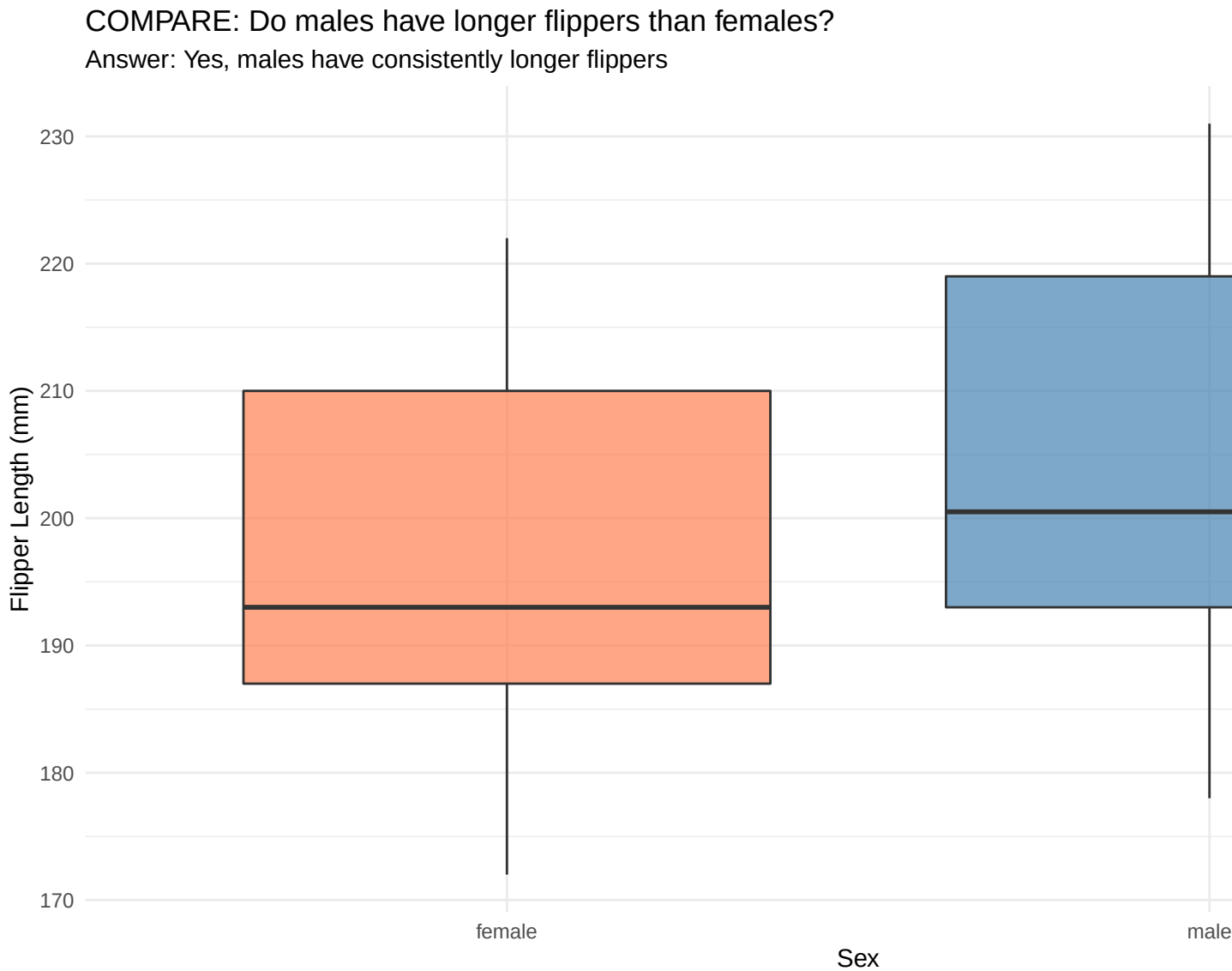
```
ggplot(penguins_clean, aes(x = island, fill = species)) +
  geom_bar(position = "dodge") +
  labs(title = "LOCATE: Where are Chinstrap penguins found?",
       subtitle = "Answer: Only on Dream island",
       x = "Island", y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```




**3. COMPARE: Do males have longer flippers?**

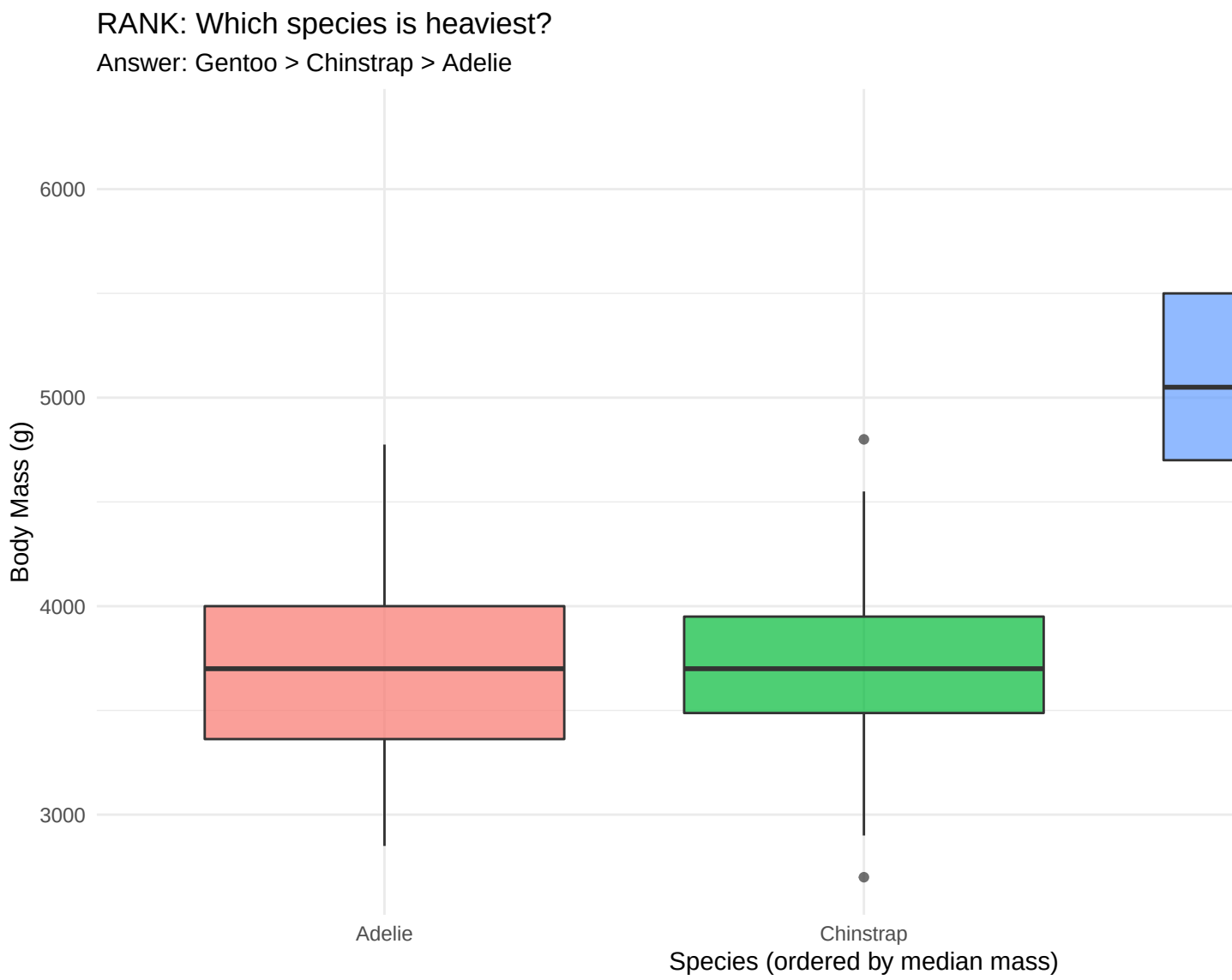
```
ggplot(penguins_clean, aes(x = sex, y = flipper_length_mm, fill = sex)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.7) +
  labs(title = "COMPARE: Do males have longer flippers than females?",
       subtitle = "Answer: Yes, males have consistently longer flippers",
```

```
      x = "Sex", y = "Flipper Length (mm)") +
  scale_fill_manual(values = c("coral", "steelblue")) +
  theme_minimal()
```



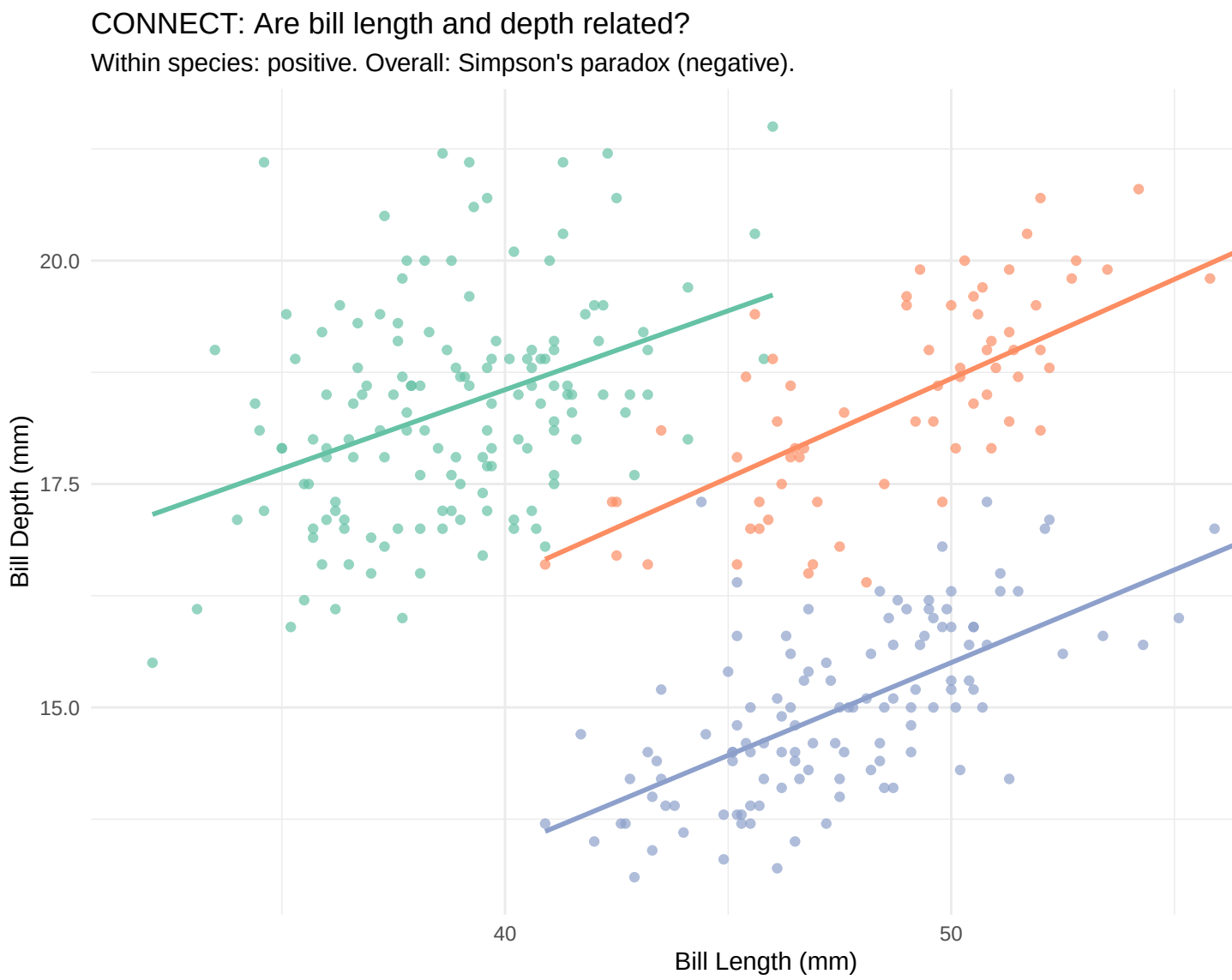

**4. RANK: Which species is heaviest?**

```
ggplot(penguins_clean, aes(x = reorder(species, body_mass_g, FUN = median),
                           y = body_mass_g, fill = species)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.7) +
  labs(title = "RANK: Which species is heaviest?",
       subtitle = "Answer: Gentoo > Chinstrap > Adelie",
       x = "Species (ordered by median mass)", y = "Body Mass (g)") +
  theme_minimal()
```




**5. CONNECT: Bill length vs bill depth?**

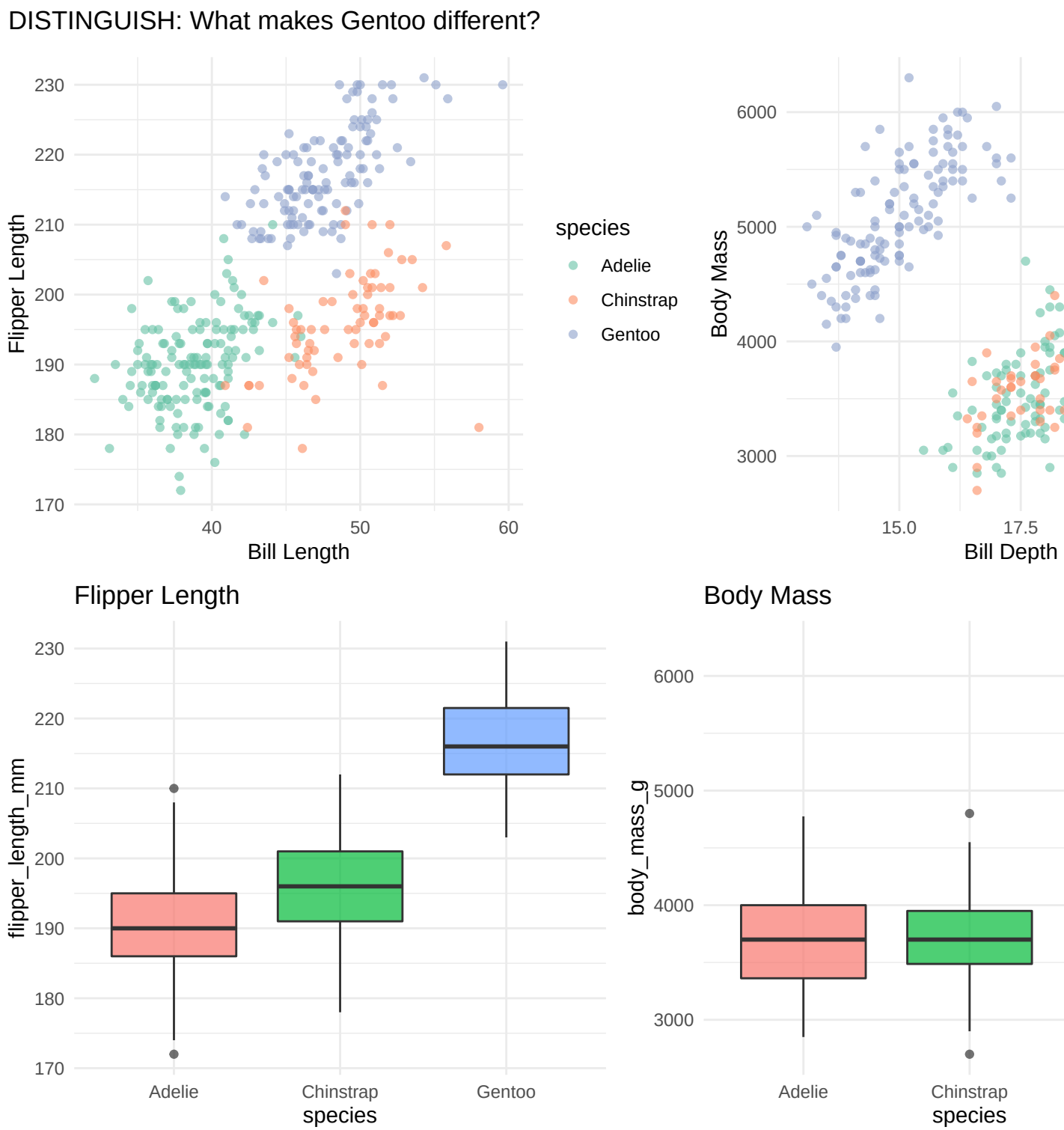
```
ggplot(penguins_clean, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "CONNECT: Are bill length and depth related?",
       subtitle = "Within species: positive. Overall: Simpson's paradox (negative).",
       x = "Bill Length (mm)", y = "Bill Depth (mm)") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```




**6. DISTINGUISH: What makes Gentoo different?**

```
p_d1 <- ggplot(penguins_clean, aes(bill_length_mm, flipper_length_mm, color = species)) +
  geom_point(alpha = 0.6) + scale_color_brewer(palette = "Set2") +
  labs(x = "Bill Length", y = "Flipper Length") + theme_minimal()

p_d2 <- ggplot(penguins_clean, aes(bill_depth_mm, body_mass_g, color = species)) +
  geom_point(alpha = 0.6) + scale_color_brewer(palette = "Set2") +
  labs(x = "Bill Depth", y = "Body Mass") + theme_minimal()

p_d3 <- ggplot(penguins_clean, aes(species, flipper_length_mm, fill = species)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.7) +
  labs(title = "Flipper Length") + theme_minimal()

p_d4 <- ggplot(penguins_clean, aes(species, body_mass_g, fill = species)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.7) +
```

```r
  labs(title = "Body Mass") + theme_minimal()

(p_d1 + p_d2) / (p_d3 + p_d4) +
  plot_annotation(title = "DISTINGUISH: What makes Gentoo different?")
```

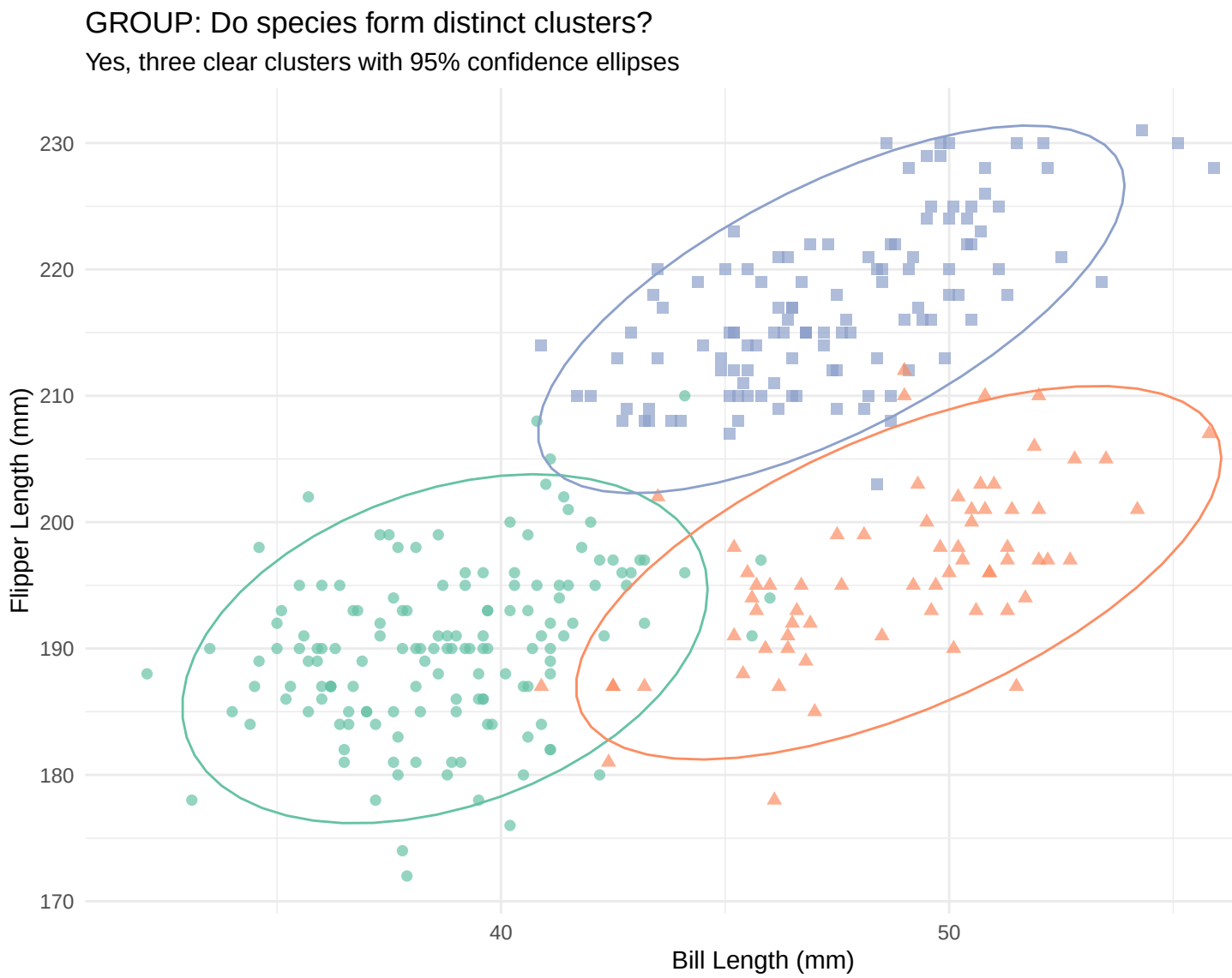



Gentoo penguins are clearly separated: larger flippers, higher body mass, and longer but shallower bills compared to Adelie and Chinstrap.

## Synoptic Questions

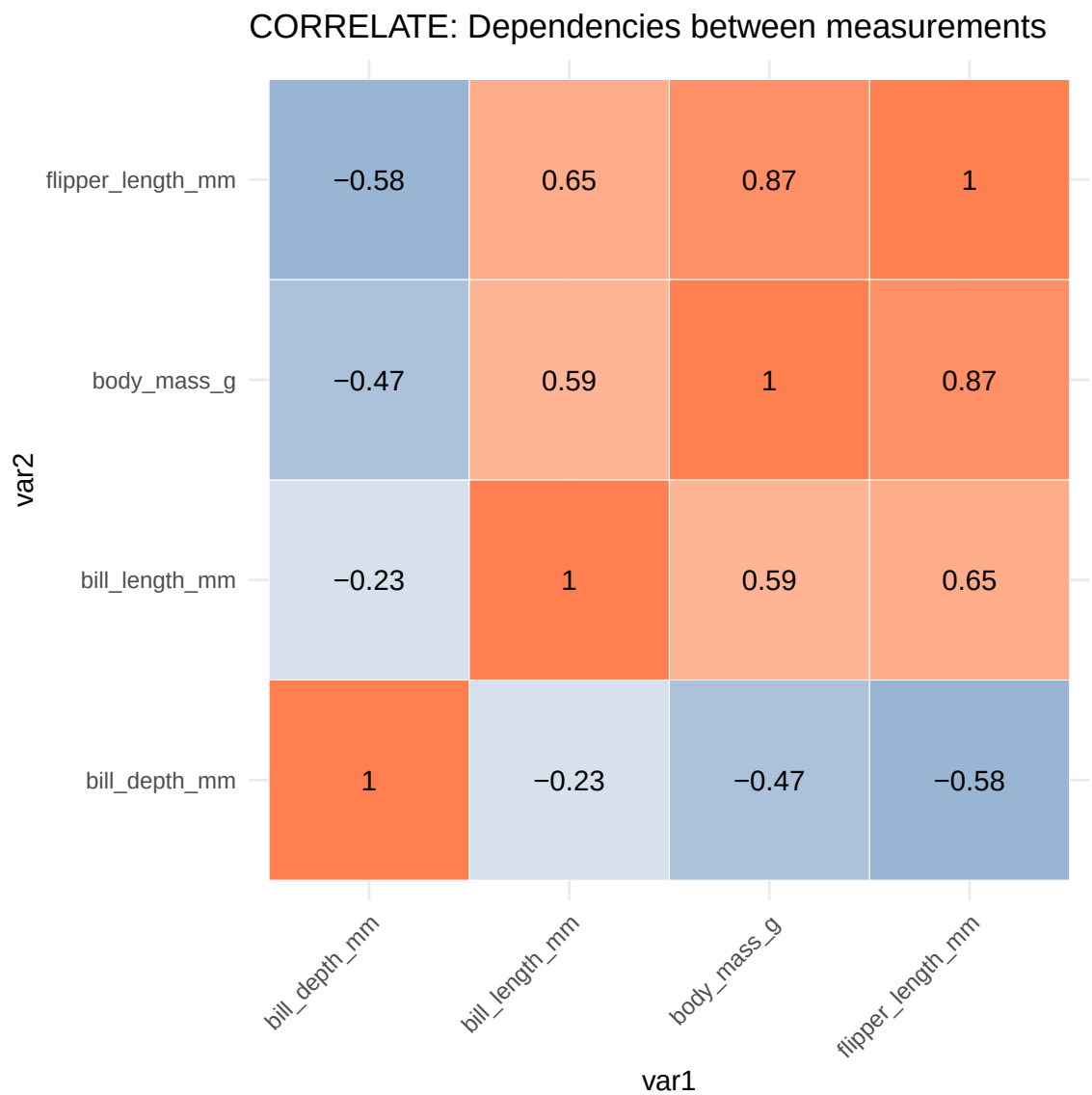
### 7. GROUP: Do species form distinct clusters?

```
ggplot(penguins_clean, aes(x = bill_length_mm, y = flipper_length_mm,
                           color = species, shape = species)) +
  geom_point(size = 2, alpha = 0.7) +
  stat_ellipse(level = 0.95) +
  labs(title = "GROUP: Do species form distinct clusters?",
       subtitle = "Yes, three clear clusters with 95% confidence ellipses",
       x = "Bill Length (mm)", y = "Flipper Length (mm)") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```



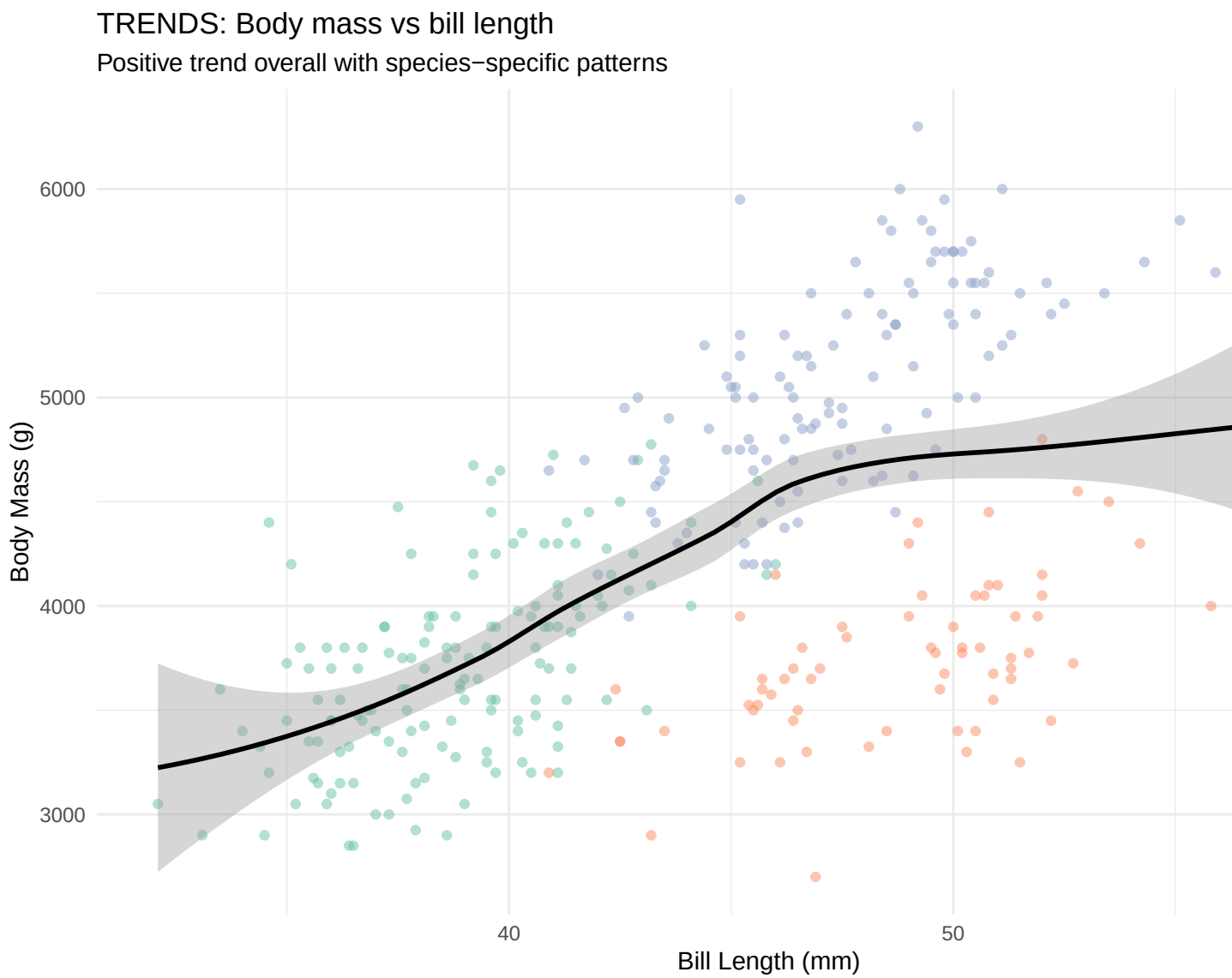
**8. CORRELATE: Dependencies between measurements?**

```r
cor_data <- penguins_clean |>
  select(bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g) |>
  cor()

cor_long <- cor_data |>
  as.data.frame() |> rownames_to_column("var1") |>
  pivot_longer(-var1, names_to = "var2", values_to = "r")

ggplot(cor_long, aes(var1, var2, fill = r)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(r, 2)), size = 4) +
  scale_fill_gradient2(low = "steelblue", mid = "white", high = "coral", midpoint = 0) +
  labs(title = "CORRELATE: Dependencies between measurements") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  coord_fixed()
```



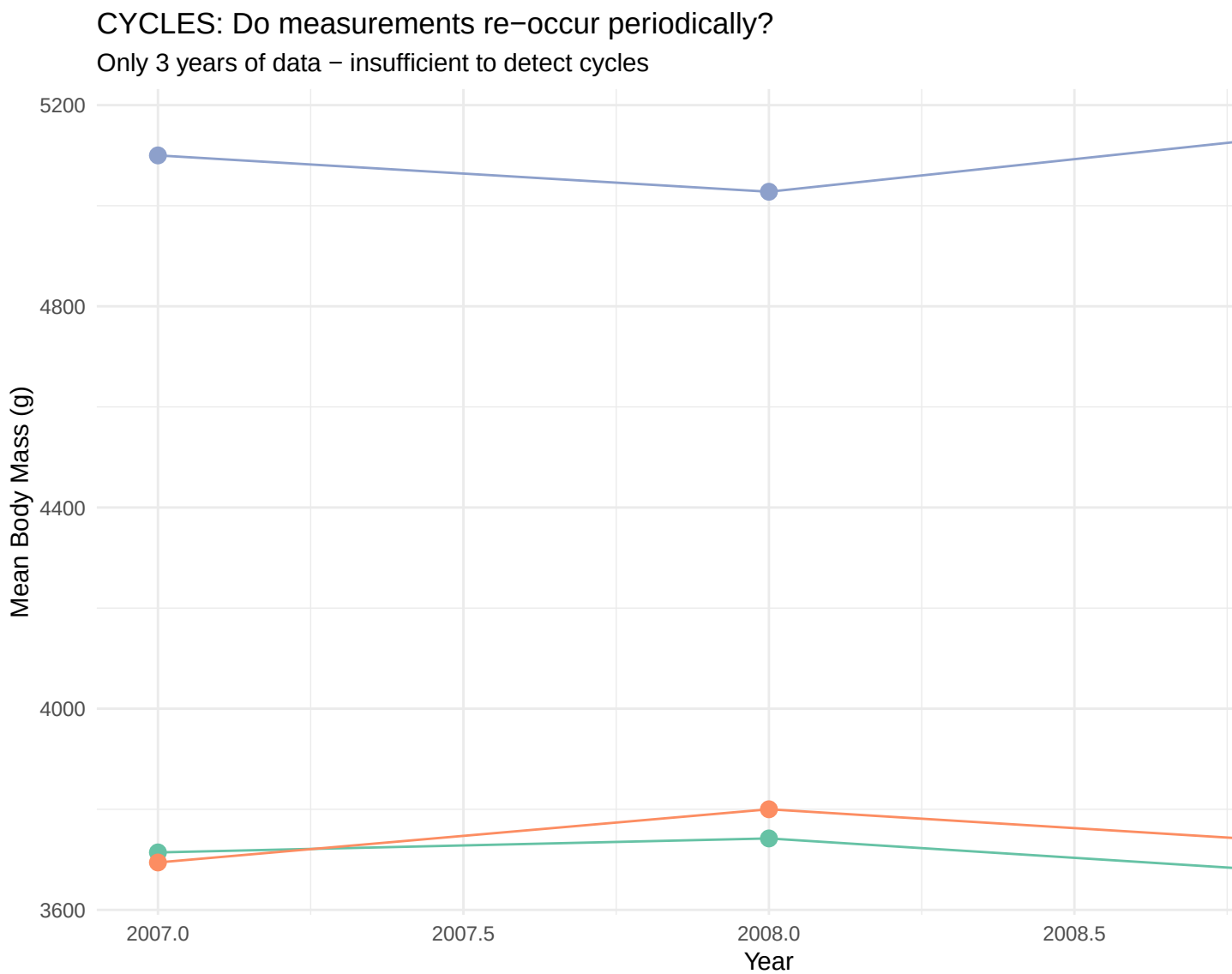

**9. TRENDS: Body mass vs bill length?**

```
ggplot(penguins_clean, aes(x = bill_length_mm, y = body_mass_g)) +
  geom_point(aes(color = species), alpha = 0.5) +
  geom_smooth(method = "loess", color = "black", se = TRUE) +
  labs(title = "TRENDS: Body mass vs bill length",
       subtitle = "Positive trend overall with species-specific patterns",
       x = "Bill Length (mm)", y = "Body Mass (g)") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```



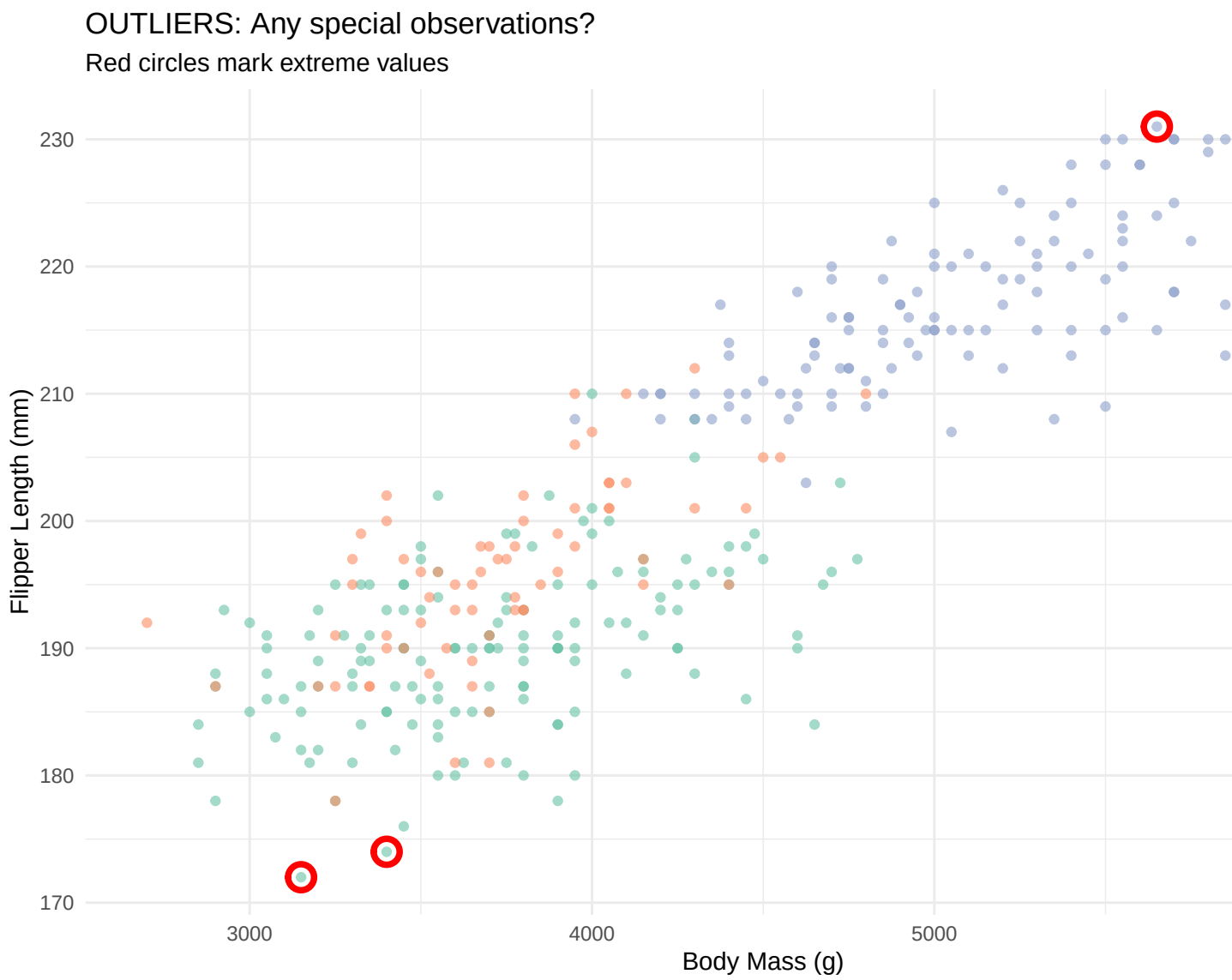

**10. CYCLES: Periodic patterns over years?**

```
penguins_by_year <- penguins |>
  drop_na() |>
  group_by(year, species) |>
  summarise(mean_mass = mean(body_mass_g), n = n(), .groups = "drop")

ggplot(penguins_by_year, aes(x = year, y = mean_mass, color = species)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(title = "CYCLES: Do measurements re-occur periodically?",
       subtitle = "Only 3 years of data - insufficient to detect cycles",
       x = "Year", y = "Mean Body Mass (g)") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```



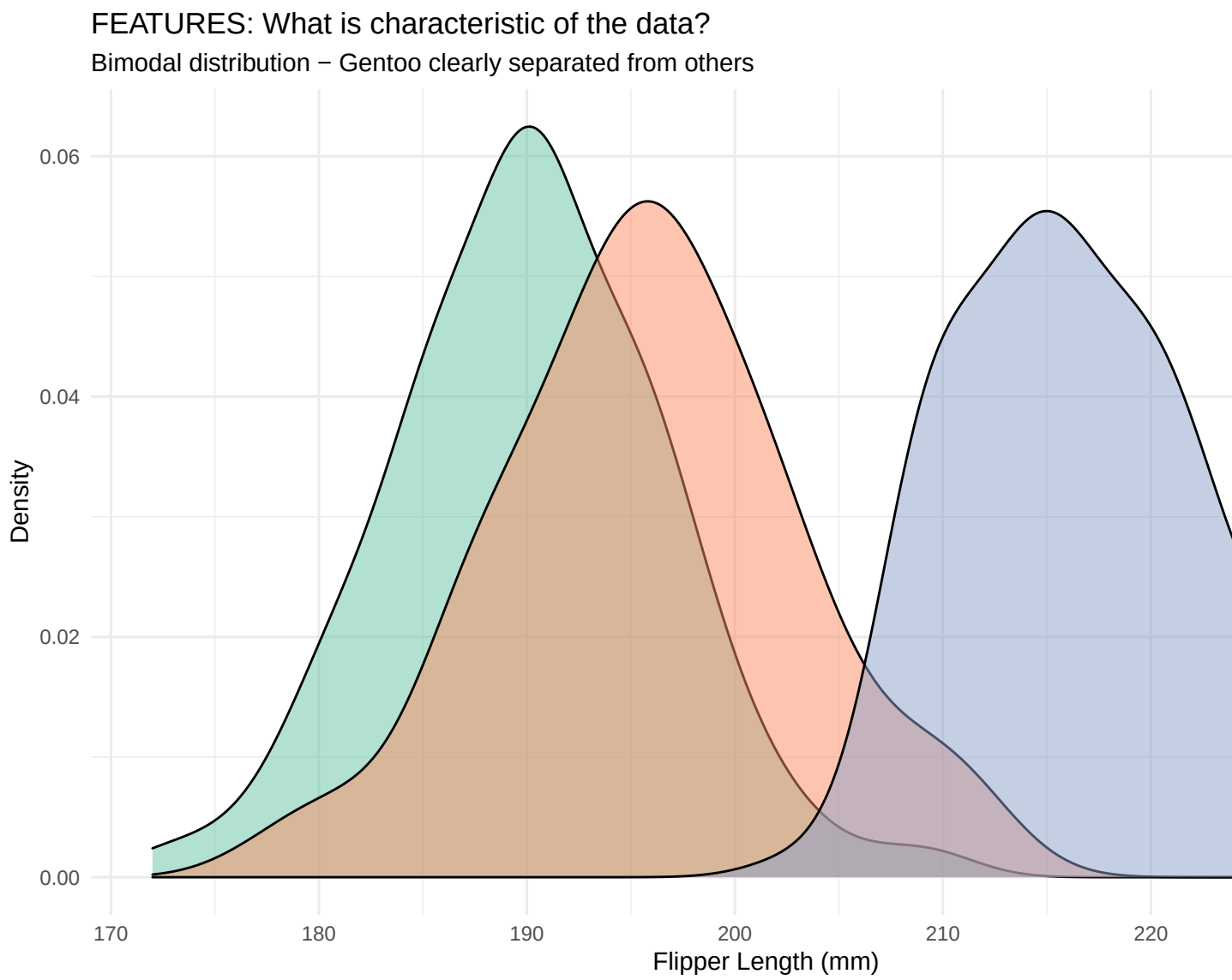

**11. OUTLIERS: Any special cases?**

```
ggplot(penguins_clean, aes(x = body_mass_g, y = flipper_length_mm)) +
  geom_point(aes(color = species), alpha = 0.6) +
  geom_point(
    data = penguins_clean |>
      filter(body_mass_g > 6000 | flipper_length_mm > 230 | flipper_length_mm < 175),
    color = "red", size = 4, shape = 1, stroke = 2
  ) +
  labs(title = "OUTLIERS: Any special observations?",
       subtitle = "Red circles mark extreme values",
       x = "Body Mass (g)", y = "Flipper Length (mm)") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```




**12. FEATURES: What characterizes the data?**

```
ggplot(penguins_clean, aes(x = flipper_length_mm, fill = species)) +
  geom_density(alpha = 0.5) +
  labs(title = "FEATURES: What is characteristic of the data?",
       subtitle = "Bimodal distribution - Gentoo clearly separated from others",
       x = "Flipper Length (mm)", y = "Density") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```




### Conclusions

- Palmer Penguins data shows **three distinct species clusters** in morphological space.
- **Simpson's paradox** appears in bill length vs depth (negative overall, positive within species).
- **Gentoo** penguins are clearly the largest species with the longest flippers.
- Only 3 years of data is insufficient for cycle detection.

```r
sessionInfo()
```

```
## R version 4.1.2 (2021-11-01)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0
##
## locale:
```

```
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] palmerpenguins_0.1.1 patchwork_1.1.1      lubridate_1.9.5
##  [4] forcats_1.0.1        stringr_1.6.0        dplyr_1.2.0
##  [7] purrr_1.2.1          readr_2.2.0          tidyr_1.3.2
## [10] tibble_3.3.1         ggplot2_3.3.6        tidyverse_2.0.0
## [13] conflicted_1.2.0
##
## loaded via a namespace (and not attached):
##  [1] RColorBrewer_1.1-3 pillar_1.11.1      compiler_4.1.2     tools_4.1.2
##  [5] digest_0.6.39      lattice_0.20-45    nlme_3.1-155       timechange_0.4.0
##  [9] evaluate_1.0.5     memoise_2.0.1      lifecycle_1.0.5    gtable_0.3.6
## [13] mgcv_1.8-39        pkgconfig_2.0.3    rlang_1.1.7        Matrix_1.4-0
## [17] cli_3.6.5          yaml_2.3.12        xfun_0.56          fastmap_1.2.0
## [21] withr_3.0.2        knitr_1.51         generics_0.1.4     vctrs_0.7.1
## [25] hms_1.1.4          grid_4.1.2         tidyselect_1.2.1   glue_1.8.0
## [29] R6_2.6.1           otel_0.2.0         rmarkdown_2.30     farver_2.1.2
## [33] tzdb_0.5.0         magrittr_2.0.4     MASS_7.3-55        splines_4.1.2
## [37] scales_1.2.0       htmltools_0.5.9    colorspace_2.1-2   labeling_0.4.3
## [41] stringi_1.8.7      munsell_0.5.0      cachem_1.1.0
```